Dataset Proposal

HackInsight 2026 | Data-to-Decision

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Problem

Smog (PM2.5) in Lahore, worst between October and February. Overnight and early-morning PM2.5 can reach very high levels (a tested sensor rose from about 43 to about 240 ug/m3 within 12 hours), but schools get little warning.

Primary user and decision

A school administrator in Lahore decides the day before whether to cancel or move outdoor activities.

Target and definitions

- **Target:** Next-day mean PM2.5 (ug/m3) per station.
- **Day definition:** A day runs midnight to midnight in PKT (UTC+5). All data is stored in UTC and converted to PKT only to define days and for display.
- **Daily mean rule:** A day counts only if at least 18 of its 24 hourly values exist.

Source 1: OpenAQ (PM2.5 from Lahore sensors)

- **Publisher:** OpenAQ, aggregating data from the providers below
- **URL:** https://openaq.org (API: https://api.openaq.org/v3)
- **Data providers:** AirGradient (station owner: The Urban Unit) and Hawanama (station owner: Pakistan Air Quality Initiative, PAQI)
- **Licence:** CC BY 4.0 (all 10 OpenAQ stations queried via GET /v3/locations/{id} on 2026-10-21T05:12:40.909576+00:00; unmodified JSON saved to docs/evidence/licences/openaq_*.json; verified in config/licence_evidence.csv; human review: TO CONFIRM BY TEAM)
- **Update frequency:** Hourly (all tested stations had updated within the last 2 hours)
- **Measured:** PM2.5 (ug/m3) at all stations. AirGradient stations also report PM1, relative humidity, temperature and PM0.3 particle count.
- **Access:** Through the OpenAQ API with a personal API key, which is stored outside the repository.

Stations (10 Discovered; 8 Used in Cleaned/Processed Layers)

#	Station Name	Provider	Reporting Since	Pipeline Status & Evidence Note
1	HBFC Society, DHA Phase 5	AirGradient	26/05/2025	Used (10,691 valid PM2.5 hrs)
2	Sandha Road	AirGradient	29/05/2025	Used (8,848 valid PM2.5 hrs)
3	Model Town	AirGradient	30/05/2025	Used (11,260 valid PM2.5 hrs)
4	College Road	AirGradient	02/06/2025	Used (9,423 valid PM2.5 hrs)
5	Pak Arab	AirGradient	19/09/2025	Used (6,950 valid PM2.5 hrs)
6	**IEEE, Punjab University, LHR**	**AirGradient**	**30/05/2025**	**Raw kept as evidence; excluded (owner unverified)**
7	Allama Iqbal Town B LHE	Hawanama	18/11/2025	Used (5,021 valid PM2.5 hrs)
8	**Model Town Link Rd B LHE**	**Hawanama**	**23/04/2026**	**Excluded: OpenAQ HTTP 500 error (config/exclusions.yaml)**
9	Forman Christian College & Chartered Univ	AirGradient	16/06/2025	Used (10,585 valid PM2.5 hrs)
10	Trinity School Lahore	Hawanama	18/11/2025	Used (2,835 valid PM2.5 hrs)

Source 2: Open-Meteo Historical Weather API (ERA5 reanalysis)

- **Publisher:** Open-Meteo
- **URL:** https://open-meteo.com (endpoint: archive-api.open-meteo.com)
- **Licence:** CC BY 4.0. Attribution: "Weather data by Open-Meteo.com". The free tier is for non-commercial use, which applies to this event.
- **Update frequency:** Hourly
- **Variables:** temperature_2m, relative_humidity_2m, wind_speed_10m, precipitation (further variables may be added)
- **Tested:** 1 to 7 Oct 2026 for Lahore, 168 hourly rows, no gaps seen, timestamps in UTC.
- **Note:** Gridded model data, not a ground station. Requests are snapped to the nearest grid cell.

Non-track source and why it is credible

Open-Meteo is not on the Air quality track list. It is built on reanalysis and national weather agency data, publishes its methods and its licence openly, and is independent of OpenAQ (a gridded weather model versus ground-level sensors).

How the sources are used in the model

- **OpenAQ PM2.5:** gives the target (next-day mean PM2.5 per station) and the history features, such as today's mean and recent hourly values.
- **Open-Meteo weather:** gives temperature, wind, humidity and rain as predictive features, because they strongly affect PM2.5.

Expected volume (Updated with Measured Counts)

10 stations discovered reporting between May 2025 and April 2026. Measured counts confirm that volume well exceeds the 50,000-row hackathon requirement: • **reports/station_validity.csv:** 75,800 valid hourly PM2.5 observations across all 9 raw stations (65,613 valid hourly PM2.5 observations across the 8 verified and used stations: HBFC Society 10,691; Sandha Road 8,848; Model Town 11,260; College Road 9,423; Pak Arab 6,950; Allama Iqbal Town 5,021; Forman Christian College 10,585; Trinity School 2,835; plus 10,187 valid raw hours for IEEE Punjab Univ kept as evidence). • **data/processed/hourly_all.parquet:** 86,373 total regularized hourly rows across the 8 used stations, with 65,877 valid non-null PM2.5 observations. • **manifest.csv:** 519 physical data artifacts tracked across raw (including the 238 raw files from 19 Oct baseline), cleaned, and processed layers. Even after excluding gaps and unverified stations, the 86,373 regularized hourly rows and >65,000 valid observations substantially exceed the 50,000-row minimum.

Why this data fits

Station data covers several areas of Lahore, so we can report errors by station and area. PM2.5 is strongly affected by weather (temperature, wind, humidity, rain), so the weather series gives the model real predictive signal. Both sources update hourly, which fits the hourly pipeline.

Impact

A school administrator gets a next-day PM2.5 estimate before deciding whether to cancel or move outdoor activities, which gives schools earlier warning of smog episodes.

Limits and risks

- Most stations are low-cost sensors, which are noisier than reference monitors.
- Only one full winter smog season (2025-26) is available, so the time-based split needs care.
- At least six of the ten stations are owned by The Urban Unit, so one feed stopping could remove many stations.
- Hawanama stations report PM2.5 only, with no humidity or temperature columns.
- Weather data is modelled on a grid, not measured at the stations.
- Outputs are information, not professional advice.

Baseline (fixed, will not change)

- **Baseline:** Tomorrow's mean PM2.5 = today's mean PM2.5 at the same station.
- **Success measure:** Model MAE at least 15% lower than baseline MAE on a later-dated test set.

Result so far: Baseline persistence MAE: 13.400 ug/m3, Model (HistGradientBoosting) MAE: 12.586 ug/m3, Improvement: 6.08% (declared 15% target not met). Read from reports/pm25_hist_gradient_boosting_metrics.json.
Note: Updated on 21 Oct 2026 with measured counts. Original version: commit 338cd28 (19 Oct 2026).